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Types, features, and effectiveness of technologies in collaborative writing for second language learning

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ABSTRACT

Technology-enhanced collaborative writing for second language learning has been receiving accumulating attention from researchers and educators over years. However, there was a lack of updated review with foci on the types and effectiveness of technologies in collaborative writing. This study conducted a systematic and in-depth review of 34 relevant empirical studies from 2009 to 2019. The results showed that wikis, Google Docs, offline word processors, Facebook, chats, and forums were the six main types of tools for collaborative writing. Technologies had overall positive effects on students' joint writing qualities, individual writing development, and perceptions of learning tasks. Advantages include prompting group interactions, helping students reflect on their work and identify errors and weakness, boosting learner motivation and confidence, facilitating writing, encouraging students to learn from others, and making the learning process enjoyable. Technologies may be ineffective when technologies were inconvenient to use, when students did not devote adequate effort to planning and reflection, and when students lacked co-responsibility for the writing. This research aims to help researchers and educators understand factors that lead to effective technology enhanced collaborative writing and assist their selection of appropriate technologies and implementation of collaborative writing in and out of class.

KEYWORDS

Collaborative writing; computer-assisted language learning; second language writing; technology-enhanced language learning

1. Introduction

Collaborative writing, which refers to the process in which a group of two or more students jointly produce a single text, has been receiving accumulating attention from researchers and educators in the field of second language (L2) acquisition over years (Coyle, Guirao, & de Larios, 2018; Li, 2018; Storch, 2011, 2013, 2019; Su & Zou, 2020). It provides students with opportunities to pool resources, apply knowledge, practise skills, have self- and peer-reflection by

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engaging them in content negotiation, feedback exchange, and joint decision making throughout the process (Lin & Maarof, 2013; Li, 2018). For the past decade, collaborative writing for L2 learning has been driven by the rapid development of digital tools and Web 2.0 applications (Chen et al., 2020; Storch, 2019; Zou, Huang, & Xie, 2019; Zou, Luo, Xie, & Hwang, 2020; Zou, 2020). Modern technologies, which enable students to co-contribute to the writing with less constraint on time or space but more assistance in the format of multimedia support and interaction approaches (Godwin-Jones, 2011; Shadiev, Hwang, & Huang, 2017), increase the accessibility, flexibility, and effectiveness of this L2 learning activity (Storch, 2013, 2019). Consequently, technology-enhanced collaborative writing (hereinafter, TECW) has been increasingly investigated and applied by researchers and educators as an effective approach to L2 acquisition (Coyle et al., 2018; Li, 2018; Zou et al., 2019).

However, little research has been conducted to systematically review the technologies that have been used to support TECW. This study therefore aims to conduct an in-depth review of the literature on TECW, with foci on the types and features of educational technologies, the effects of these technologies on students' learning outcomes and perceptions of TECW. The results are expected to help researchers and educators understand factors that lead to effective use of technologies to support TECW and assist their selection of appropriate technologies and implementation of TECW in and out of class. The research questions are listed as follows.

1. What educational technologies were used to support TECW?
2. How did educational technologies affect students' learning outcomes and perceptions of TECW?

2. Literature review

2.1. Collaborative writing

Collaborative writing, different from group-planning and peer-feedback activities, requires writers' co-working and interactions "in all stages of the writing process" rather than in a part or parts of it (Storch, 2019, p.40). It also does not allow any division of labour (Storch, 2013, 2019).

Storch has made an abundance of contributions to review studies of collaborative writing. In (2011), she reviewed previous studies on collaborative writing with foci on the cognitive and socio-cultural aspects. Upon her analyses and categorisation, the researcher identified a series of factors in TECW that might influence language learning outcomes, such as types of writing task, L2 proficiency levels of students, types of educational technologies, and patterns of interaction.

In (2019), Storch combed through the most important empirical studies on collaborative writing published from 1994 to 2017 and identified two stages of the research timeline. In the first stage (from 1994 to 2009), most researchers

implemented collaborative writing activities using traditional methods in which students wrote in pairs, on paper, face-to-face and in classrooms. In the second stage (from 2009 to 2017), most collaborative writing experiments were computer-mediated with Internet access. In the review, Storch argued that the predominant foci of the studies on collaborative learning lied on learners' behaviours during the process of writing and interaction and their language learning outcomes of this activity.

2.2. Technology enhanced language learning

Technology-enhanced language learning has long been a focus of interest in L2 education, on which many researchers have conducted review studies. For example, Hung, Yang, Hwang, Chu, and Wang (2018) reviewed 50 studies on digital game-based in language learning published from 2007 to 2016 from the aspects of research methods, digital games, target languages and language learners, as well as student learning outcomes. They found that custom-built, computer-based immersive games, notably massively multiplayer online role-playing games, were investigated most frequently in the context of language education. English as the target language and university students as target language learners were investigated most frequently. Fairly convincing evidence was found supporting the overall positive effects of digital game-based learning on learners' affective states and language acquisition.

Hwang and Fu (2019) overviewed the development of collaborative mobile learning in various disciplines by reviewing 90 relevant studies published from 2007 to 2016. Their coding scheme composed of three major categories, specifically, methodology, languages and learner types, and learning outcomes. The results of their review identified the increasing investigation and application of collaborative mobile learning in the context of language learning. English as second language and secondary and university students as the target language learners were investigated most frequently.

Fang, Hwang, and Chang (2019) conducted a systematic review of 33 studies on open online language learning published from 2009 to 2018 from the aspects of research design, sample size, learning devices, target languages, language skills, the learners' education level, the learners' language proficiency level, learning methods, and the foci of investigation. The results showed that most researches in this field were quantitative empirical studies or mixed research studies. English as the target language and reading and writing as the target aspects of language learning were investigated most frequently. Most open online language learning adopted self-directed learning, blended learning, and peer review method. Learner motivation, interest, and satisfaction in online course were vastly discussed. The researchers suggested that future research may investigate the effectiveness of online language learning that adopted other learning methods, such as collaborative learning, flipped classrooms, adaptive learning, and thematic-oriented learning.

2.3. Technology enhanced collaborative writing (TECW)

TECW refers to the collaborative writing activity using digital tools and technologies (Li, 2018; Storch, 2019). With the aid of technologies, students could have easier and more accessible “creation and sharing of texts” and higher efficiency of writing and communication than what they used to have in traditional, face-to-face, paper-based collaborative writing (Storch, 2019, p.40). For example, Godwin-Jones (2003) reported a list of technologies that can be useful in TECW, ranging from asynchronous and synchronous tools, such as Wimba and MSN Messenger, to Web 2.0 platforms/applications, such as blogs, RSS and wikis. In the review, MSN Messenger and RSS were listed as potential technologies that could be used in TECW.

Li (2018) focused on computer-mediated collaborative writing and conducted a systematic review of 21 relevant articles published from 2008 to 2017, with foci on context and participants, theoretical/pedagogical framework, writing tasks, used technology, and foci of investigations. The results showed that most studies conducted in the tertiary L2 context, that socio-cultural theory was the most frequently applied framework, and that most studies investigated the collaborative writing in small group tasks. Wiki, Google Doc, and chat were the technologies used most frequently in computer-mediated collaborative writing. The researcher also identified three major research strands of these studies, namely, interaction/writing process, writing products/outcome, and students’ perceptions.

2.4. Limitations of previous review studies

Considering the importance and popularity of the application of technologies for language education (Shadiev & Yang, 2020; Zou et al., 2020; Chen et al., 2020), it is necessary to conduct a systematic review of the literature on TECW with foci on the types and features of technology used therein for comprehensive understanding of the field. However, little has been done, so far as we know. Godwin-Jones (2003) listed various types of technologies that were potentially useful in collaborative learning but did neither substantiate the list nor specify the use and features of these technologies with empirical evidence from the literature on TECW. Storch (2011, 2019) focused on collaborative writing in general, so they did not discuss the educational technologies. Li (2018) reviewed 21 studies on computer-mediated collaborative writing and found wiki and Google Doc the most frequently used but did not specify the features or applications of the technologies. However, to help researchers and educators select appropriate technologies and features in TECW, it is valuable to provide an evidence-based list of types and features of technology that were useful in TECW.

Moreover, although the previous review studies had identified students’ outcomes and perceptions of TECW, they did not investigate effects of technologies on those aspects (e.g., Li, 2018; Storch, 2011, 2019). The previous

studies reported that the affordances of technologies mediated the implementation and outcomes of TECW (Li & Kim, 2016; Wong, Chen, Chai, Chin, & Gao, 2011; Woo, Chu, Ho, & Li, 2011). Thus, to facilitate researchers and educators to use technologies and features effectively, it is helpful to review the literature with foci on how technologies affected students' outcomes and how students perceived technologies.

In addition, the previous review studies in this field have yet covered the publications after 2017, leaving the latest progress and newest situations un-reviewed. Considering the rapid development of technology-enhanced language learning methods, the technologies that used to be innovative before 2017 may have seemed out-dated at the present time (Godwin-Jones, 2011; Shadiev & Yang, 2020). Thus, it was valuable to conduct a timely review on TECW.

3. Research method

We conducted the review employing a three-step process: search, selection, and data analysis, following Zou et al. (2019, 2020) (see Figure 1). The search was conducted in Web of Science, with “2009–2019” as the time-span, “English” as the language, and “article” as the required document type. Based on Storch (2011, 2019) and Li (2018), we selected the keywords as follows, (a) “collaborative

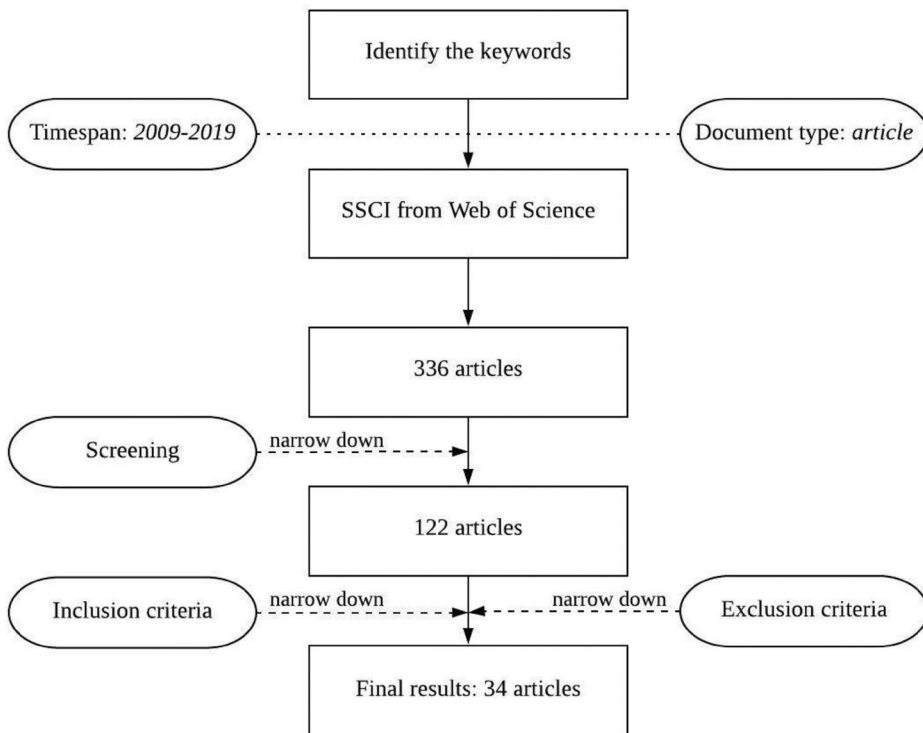


Figure 1. Process and methods of data search, selection, and collection.

writing” or “write collaboratively” or “collaboratively write” or “collaboratively wrote” or “collaborative written text” or “learn collaboratively through writing” or “collaborative learning,” (b) “language learning” or “learn language” or “language education” or “language acquisition” or “teach language” or “language teaching” or “English learning” or “English teaching” or “English education,” and (c) “second language” or “foreign language” or “L2” or “SL” or “FL” or “ESL” or “EFL,” with the AND operators between. “SL,” “FL,” “ESL,” and “EFL” were respectively the acronyms of “second language,” “foreign language,” “English as second language,” and “English as foreign language.” The time-span of this review was the period of one decade from 2009 to 2019 because we found that many review papers in the field of technology-assisted language learning had a decade as time span (e.g., Hung et al., 2018; Shadiev et al., 2017).

We decided to only include the articles that were published in Social Sciences Citation Index (SSCI) journals in this review, though conference papers, book chapters, and non-SSCI journal articles were covered at the prior stage. This is because we found specific information of the educational technologies and detail explanations of the research findings essential for in-depth analysis, while they were not always specified in conference papers, book chapters, and non-SSCI journal articles.

The search generated a total of 336 articles. The two authors of this study firstly excluded 214 irrelevant articles from the review list by screening the titles and abstracts and then examined the remaining 122 articles according to the following inclusion and exclusion criteria. The first inclusion criterion was that the research topic should be TECW for L2 learning, based on which, 66 articles were excluded. The research findings should also be supported with empirical data, based on which, 11 articles were excluded. Concerning the exclusion criteria, research on teacher training was excluded as our review foci were on students’ learning. This criterion excluded eight articles. The studies on students with special needs were also excluded, considering the discrepancy between the language education for general students and that for students with special needs (Scott & Windsor, 2000). This criterion excluded three articles. A total of 34 articles were finalized for the review, as listed in [Appendix A](#).

These articles were reviewed from three dimensions.

1. Technology (i.e., the digital tools, applications, and platforms used by students to perform TECW). The main categories included wiki, Google Doc, chat, and blog, according to Godwin-Jones (2003) and Li (2018).
2. Effects of technology on students’ outcomes of TECW. This category was analysed from three aspects. One concerned the types of outcomes, which, adapted from Li’s (2018) category, fell into students’ joint writing qualities and individual writing development. The second aspect concerned the results of the investigation of students’ writing outcomes and included three sub-categories, namely, positive, negative, and neutral. The third aspect concerned the reasons for those results.

3. Effects of technology on students' perceptions of TECW. This category included two aspects. One concerned the results of the investigations of students' perceptions of TECW, falling into three sub-categories that were positive, negative, and neutral. The other concerned the technical reasons for those results.

The two researchers of this study analysed five articles together first to generate the initial codes and reach the agreement on coding methods and then analysed remaining articles independently. The coding results were compared afterwards, in which several differences were identified. The differences were then resolved via discussions between the researchers and consultation with a third rater who was an expert in this field. At the end of the coding, satisfactory agreement was achieved (Pearson's $r = 0.90$).

4. Results

4.1. Technology

This review identified 11 types of technology used in the 34 TECW studies (see Figure 2). Wikis were used in 21 articles (62%), Google Doc in six articles (18%),

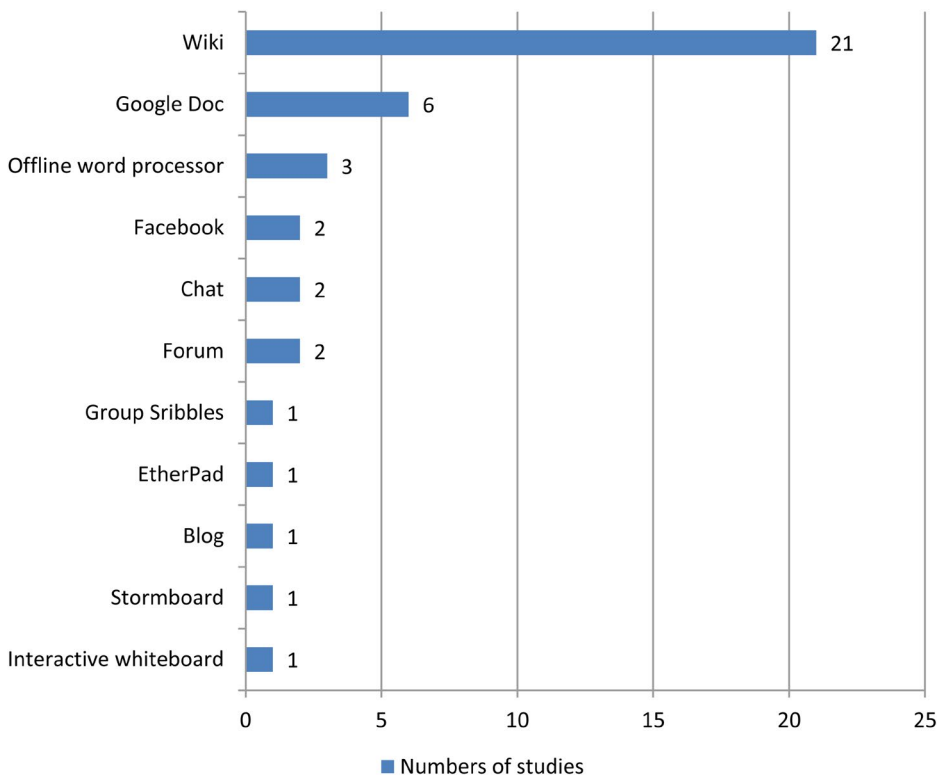


Figure 2. Types of technology for TECW.

offline word processors in three articles (9%), Facebook in two articles (6%), chats in two articles (6%), and forum in two articles (6%). Group Scribbles (i.e., a web-based workspace for visualization of idea exchange and content negotiation) (Wen, Looi, & Chen, 2015), EtherPad (i.e., an web-based collaborative word processor) (Yeh, 2014), blog (sina.com) (Zhang et al., 2014), Stormboard (i.e., a web-based conferencing tool) (Levrai & Bolster, 2019), and interactive whiteboard (Wen et al., 2015) was respectively used in one article (3%). Many articles applied more than one type of technology (e.g., Levrai & Bolster, 2019; Razak & Saeed, 2014), more details of which are presented in [Appendix B](#).

4.1.1. Wikis

Wikis were regarded as “the most relevant technology for collaborative writing” by Chao and Lo (2011, p. 378). Their use was usually computer mediated (Elola & Oskoz, 2010; Woo, Chu & Li, 2013). Examples of wikis include Wikispaces (wikispaces.com) (e.g., in Chao & Lo, 2011; Li & Zhu, 2013), PBworks (also named as PBwiki) (pbworks.com/academic.wiki) (e.g., in Elola & Oskoz, 2010), and Wetpaint (wetpaintart.com) (e.g., in Lin & Yang, 2011).

4.1.1.1. Wiki features that were useful for students. Wikis included six features that were useful for students in TECW.

1. Convenient writing and revising. Wikis presented neat organisation of multiple writing pages (Wong et al., 2011) and allowed students to go directly into the target page and work on it at the target point (Bradley, Lindström, & Rystedt, 2010). A plentiful of editing functions were provided that afforded drafting, editing, and revising (Wang, 2014, 2015). Multiple students could use those functions asynchronously (Wong et al., 2011). For example, wikis enabled the practice of revising to be easy and efficient. Via “History” module of wikis, students had a lucid view of all versions of texts in which changes in text of different types (e.g., addition, deletion) were conspicuous by different colours (Bradley et al., 2010; Li, 2013; Wong et al., 2011). Non-linear navigation for text was also provided in wikis, so students could review the changes following the organisation of the text rather than the conventional chronological order (Chao & Lo, 2011). When they spotted words/phrases that they did not understand, students could place a question mark beside the target points to request for explanations (Wong et al., 2011). Wong et al. (2011) took full advantages of the convenient writing of wikis. Their participants used editing functions of Wikispaces to carry out a series of collaborative writing practices, namely, “word/phrase pooling,” “sentence making,” “paragraph writing,” “outlining,” and “essay writing.”
2. Convenient group interactions. The “Discussion” module of wikis afforded asynchronous commentary and communication among group members (Chao & Lo, 2011; Woo et al., 2011), which enabled

students to exchange ideas and negotiate contents at different stages of writing (Li, 2013). Zorko (2009) reported that the students demonstrated easy and enthusiastic participation in group communication and share information, resources, and knowledge in “Discussion.”

3. User-friendliness. The writing and discussion functions of wikis were easy-to-learn and easy-to-use (Wang, 2014, 2015), because they were simple and similar to conventional editing and commentary styles (Chao & Lo, 2011). Evidence was found in Ducate, Anderson, and Moreno (2011) who reported that very few of their participants listed “technical difficulties” as a problem in TECW.
4. Learning materials and tools. Wiki pages contained a side bar where students could find links to learning resources and tools that were placed by their teachers, such as instruction for assignments (in Bradley et al., 2010), spell check tool (in Woo et al., 2011), and guidelines (in Zorko, 2009). Zorko (2009) found that the participants used the resources attached in the side bar very frequently, especially the Writing Reports resource that helped them check the elements and structure of writing. The researcher also identified the positive correlation between the availability of resources in wikis and students’ sense of responsibility for their writing.
5. Multimedia technologies. Wikis supported the application of multimedia technologies in collaborative writing, including image, audio, and video (Woo et al., 2011; Woo, Chu, & Li, 2013). In Woo et al. (2013), for instance, the students were required to use PBworks to co-construct two texts and add photos and graphics thereto.
6. Personalised scaffolding. Wikis provided templates that could scaffold students’ writing at different stages (Li, 2013). Those templates also allowed personalisation setting so as to fit students’ individual needs and proficiency levels (Zorko, 2009).

4.1.1.2. Wiki features that were useful for researchers and educators. Wikis were useful for researchers and educators in TECW due to the following four features.

1. Automatic saving, organising, and analysis of historical records of the process of TECW. Wikis automatically archived the entire process of group interactions/writing (Li & Zhu, 2016, 2017), coded all students’ contributions in different colours (Bradley et al., 2010), and organised them collectively (Wang, 2015), which provided researchers and educators with a data source for their investigation of students’ behaviours during TECW (Li & Zhu, 2016; Wong et al., 2011). For example, Li (2013) examined the communication of one collaborative writing group by analysing their “Discussion” records. The researcher printed a total of 36 discussion posts regarding writing tasks, numbered the posts sequentially, and read the posts to derive categories of them. Based on the structured analysis, Li identified five thematic categories of group

communication in wiki writing, namely, content discussion, social talk, task management, technical communication, and language negotiation. To investigate the students' writing and revision behaviours, Li also tracked their writing changes in "History" module, saved these data in a Word document, numbered them in a chronological order, and coded them into five categories, specifically, addition, deletion, rephrasing, reordering, and correction. The results showed that addition and correction were the types of revision that occurred most frequently in TECW, followed by rephrasing, reordering, and deletion. Moreover, wikis afforded basic data analyses, such as charting the frequency of students' visits and postings and providing an overall picture of students' contributions and interactions in TECW (Woo et al., 2011).

2. Group setting. Wikis afforded easy setting-up of class wikis and wiki pages for different groups and neatly organised them in the navigation bar (Ayđın & Yildız, 2014; Chao & Lo, 2011), so researchers and teachers could manage the writing activities of multiple groups efficiently (Wang, 2014, 2015). The group setting could be modified by privacy setting. Wiki groups could be entirely open wherein both members and external visitors were allowed to view and post information (Bradley et al., 2010), which was perceived beneficial by many students for enabling them to compare their writing with that of other groups and learn from each other (Zorko, 2009). Wiki groups could also be password-protected and accessible only for members, which may help students develop team spirit and co-responsibility for their writing (Kessler, 2009).
3. Feedback giving. Due to the automatic record saving and analysing system of wikis, researchers and teachers were able to assess students' performances in collaboration and development in writing during the process, rather than, like in the traditional group work, at the end when the task was finished, so they could spot students' problems in-time and provide them with immediate and accurate feedback (Woo et al., 2011, 2013; Zorko, 2009). Feedback included scaffolding (Woo et al., 2011, 2013), encouragement and comment (Zorko, 2009), and instructional support about the content, structure, organization, and form of writing (Elola & Oskoz, 2010). Zorko (2009) suggested that wikis help teachers communicate more frequently and efficiently with their students.
4. Simplicity to be incorporated into a course. Wang (2014, 2015) reported that they found the use of wikis could be integrated into language courses smoothly and easily.

4.1.2. Google docs and EtherPad

Google Docs and EtherPad were web-based collaborative word processors. Their uses were usually computer mediated (e.g., in Abrams, 2016, 2019; Yeh, 2014).

4.1.2.1. Google docs and EtherPad features that were useful for students. Google Docs and EtherPad had five useful features for students in TECW.

1. Convenient group interactions. Google Docs and EtherPad afforded convenient commentary and discussion functions. Different from wikis which only support asynchronous communication, they enabled multiple students to synchronously engage in interactions by providing them with a real-time text-chat box in the right column (Cho, 2017; Yeh, 2014).
2. Convenient writing and revising. Similar to wikis, Google Docs and EtherPad offered a variety of functions that allowed multiple users to practice drafting, editing, and revising simultaneously (Bikowski & Vithanage, 2016; Kessler, Bikowski, & Boggs, 2012; Yeh, 2014). The newest versions of writing were updated quickly (Kessler et al., 2012), and all versions could be tracked, edited, and reverted (Bikowski & Vithanage, 2016; Yeh, 2014). Different from that in wikis, the navigation for text in Google Doc was linear (Abrams, 2016; Yeh, 2014). Kessler et al. (2012) reported that students tended to be more willing to express their opinions and explore new ideas in Google Doc writing than usual because they found all their changes in text could be undone and viewed by other members.
3. Availability of special characters, tables, and hyperlinks. Abrams (2016, 2019) emphasised that Google Docs allowed the use of special characters which were needed in many languages, such as German in their cases. Google Docs also supported the use of tables, images, and hyperlinks, which made many types of writing tasks available.
4. User-friendliness. Compared to wikis, students were generally more familiar with Google Docs (Abrams, 2016, 2019). Moreover, Abrams (2019) argued that Google Docs were particularly steady and “more readily available” than wikis (p. 33), so most students could use it easily and smoothly. EtherPad was also easy-to-learn and -use because it shared a lot of features with Google Docs (Yeh, 2014).
5. Multimedia technologies. Google Docs supported the use of pictures in collaborative writing (Abrams, 2019).

4.1.2.2. Google docs and EtherPad features that were useful for researchers and educators. Researchers and educators may found Google Docs and EtherPad useful in TECW due to the following two features.

1. Automatic saving and organising of historical records of the process of TECW. Similar to wikis, Google Docs and EtherPad automatically saved all versions of writing, transcripts of text-chats, and students’ writing action logs, stored them collectively, and coded individuals’

- contributions with different colours (Abrams, 2019; Yeh, 2014). This was helpful for researchers and teachers to monitor, track, and assess students' writing and communication process (Abrams, 2016; Bikowski & Vithanage, 2016; Kessler et al., 2012; Yeh, 2014). For example, Yeh (2014) analysed all transcripts of group interactions recorded in EtherPad chatbox, counted the numbers of students' texts in each group, and reached the collective mean of group collaborative dialogues. Based on the analyses, the researcher classified the participant groups into three categories: (a) groups with high proportions of collaborative dialogues that were in the top 33% of the collective mean, (b) groups with median proportions of collaborative dialogues that were in the median 34%, and (c) groups with low proportions of collaborative dialogues that were in the bottom 33%. In addition, Google Doc had the time-stamp function that recorded how much time students actually spent on various stages of writing (Abrams, 2019). By analysing the data of time-stamp, Abrams (2019) found that some groups did not start composing immediately after their brainstorming but spent a long time on writing preparation.
2. Group setting. Abrams (2019) reported that both teachers and students could easily set up groups in Google Docs writing environment, similar to that in wikis.

4.1.3. Offline word processors

Different from wikis and Google Docs, which require internet access, some collaborative writing activities were implemented with offline word processors. In such contexts, students had face-to-face interactions and took turns typing up their jointly produced texts in one computer using word processors. For example, Van Steendam, Rijlaarsdam, Van den Bergh, and Sercu (2014) and Wang (2015) used Microsoft Word. Compared to paper-based collaborative writing, the use of offline word processors made students' practice of writing more easily and efficiently and may thereby increase students' effort for communications and collaborations in groups.

4.1.4. Facebook and blog

Facebook and blog enabled users to generate and exchange information by posting, liking, and reposting comments and messages (Razak & Saeed, 2014; Selcuk, Jones, & Vonkova, 2019; Zhang et al., 2014). In TECW, they were used mainly for information sharing and group interactions, while students produced text through other types of technology, such as wikis (e.g., in Razak & Saeed, 2014).

4.1.4.1. Facebook and blog features that were useful for students. As supportive tools for TECW, Facebook and blogs had four useful features for students.

1. Learning materials. Facebook could be used as a virtual repertoire where students shared useful learning resources and instructions by posting and post reading (Razak & Saeed, 2014; Selcuk et al., 2019). For example, in Razak and Saeed's (2014) study, students received daily posts as resources for learning L2 reading, grammar, vocabulary, and writing. Instructions and tips on collaborative writing strategies were also shared, such as the group communication skills and the information about goals and practices of writing. Moreover, Facebook provided students with reminders of the newest posts of their teachers and peers, so they would not miss important information (Selcuk et al., 2019).
2. Convenient group interactions. Facebook and blogs allowed students to comment on others' posts (Razak & Saeed, 2014; Zhang et al., 2014). For example, in Zhang et al.'s (2014) study, students were required to post their drafts of writing in their blogs, and their group members were required to discuss and help improve the drafts by commenting on the posts. In addition, "Chat" and "Discussion Board" of Facebook afforded multiple students to simultaneously engage in group interactions where students could negotiate content, carry out their shared practices, share links, upload and download files (Razak & Saeed, 2014; Selcuk et al., 2019).
3. Availability on various devices, such as computers and smartphones. Different from most other types of useful technology in TECW, which were only computer-mediated, Facebook and blog (sina.com) were both computer-mediated and mobile-based. Selcuk et al.'s (2019) participants reported that they preferred mobile devices (especially smartphones) to laptops and desktop computers, so they engaged in collaborative writing more frequently using mobile devices than using computers.
4. User-friendliness. Facebook and blog (sina.com) enjoyed huge user population, so most L2 students had been familiar with them before TECW projects and showed great competence and interest in using them (Selcuk et al., 2019; Zhang et al., 2014). Selcuk et al. (2019) reported that none of their participants voiced any difficulty in using Facebook for TECW.

4.1.4.2. Facebook feature that was useful for researchers and educators. Facebook was featured of automatic saving of historical records of the communication process, which was useful for researchers and educators to implement TECW. All posts and comments generated by these posts as well as all dialogues in "Chat" and "Discussion Board" were achieved and traceable, which were valuable for

the assessment and investigation of group interactive patterns and individuals' contributions in interactions (Razak & Saeed, 2014; Selcuk et al., 2019). For example, to investigate their students' participation in group interactions, Razak and Saeed (2014) counted the numbers of their posts, reposts, and comments in Facebook, and as a result identified their overall active participation in the activities. Selcuk et al. (2019) investigated the transcripts in "Chat" and "Discussion Board," numbered them in different Word documents, coded them with different colours, and clustered and organised the codes into broader categories. The results of analyses helped them identify the roles and behaviours of group leaders in TECW.

4.1.5. Chats

Chats referred to the web-based real-time communication systems (Elola & Oskoz, 2010). Cho (2017) used Skype chat, and Elola and Oskoz (2010) used a university-developed chat system, both of which supported text-chats and voice-chats. In the literature, researchers employed chats to facilitate students' interactions, while students practiced writing using other types of technology, such as wikis (e.g., in Elola & Oskoz, 2010) and Google Docs (e.g., in Cho, 2017).

4.1.5.1. Chat features that were useful for students. Chats were featured by convenient group interactions, which were highly self-paced and accessible (Elola & Oskoz, 2010). Students could access word processors from multiple sites to co-work on the writing while communicating through chats (Cho, 2017; Elola & Oskoz, 2010).

4.1.5.2. Chat features that were useful for researchers and educators. Chats afforded automatic saving of historical records of the communication process of TECW. Similar to Google Docs text-chat and Facebook "Chat," all students' contributions in discussion were automatically archived by chats, serving as a data source for researchers and teachers to investigate group interactions (Cho, 2017; Elola & Oskoz, 2010). For example, Cho (2017) coded all students' posts in chats to identify in what types of episodes students engaged (i.e., the segment of group conversations on the same topic). The results showed that there were three categories of episode types, namely, (a) on-task talk, which was on the content discussion, formatting, and editing of the summaries, (b) about-task talk, which was on labour division, decision making, and performance monitoring, and (c) off-task talk, which was on topics irrelevant to the writing tasks.

4.1.6. Forums

Forums were useful in TECW by supporting the uploading, downloading, and commenting of writing (Mohamadi, 2018; Levrai & Bolster, 2019). Examples

of forums include the Moodle forum used in Levrai and Bolster (2019) and an e-writing forum website (e-writingforum.ir) used in Mohamadi (2018).

4.1.6.1. Forum features that were useful for students. There were two useful forum features for students in TECW.

1. Convenient group interactions. Students could leave in-line comments on texts in forums to negotiate content and improve language accuracy (Levrai & Bolster, 2019; Mohamadi, 2018). Forums also provided discussion tools through which multiple students simultaneously engaged in idea sharing, review changing, and feedback gathering (Mohamadi, 2018).
2. Convenient revising. In e-writing forums, students' drafts could be uploaded thereto and open for group members to revise and make changes (Mohamadi, 2018). All the changes were traceable, and the students who uploaded the drafts could decide whether they accepted or rejected these changes (Mohamadi, 2018).

4.1.6.2. Forum feature that was useful for researchers and educators. Forums were featured by feedback giving. By in-line commentary, researchers and teachers could provide accurate suggestions, instructions, and encouragements to students' writing (Levrai & Bolster, 2019; Mohamadi, 2018). For example, in Levrai and Bolster (2019), teachers read multiple drafts of students' co-writing and provided feedback focusing on high-order concerns, such as clarity of thesis, organization, and use of referenced support.

4.1.7. Group scribbles

Group Scribbles (groupscribbles.sri) used in Wen et al. (2015) was a useful tool for students in TECW, featured by convenient writing and group interactions. The interface of this software consisted of two panes of windows, specifically, the lower pane (i.e., private board) and the upper pane (i.e., the public board). The private board was only visible for the individual users who were free to draw or type whatever ideas that came into their mind thereon. When they wanted to share their ideas, they could drag their works from the private board to the public board which was visible to all group members. If students wanted to use or edit the works posted by other members, they could just take them from the public board to their private board. The combination of the private board and the public board enabled students to proactively engage in group interactions and smoothly transfer between individual and collaborative works of text production and idea generation.

Group Scribbles was also useful for researchers and teachers because of its group setting feature. Researchers and teachers could group their students in Group Scribbles and decide whether these students were allowed to browse the public boards of other groups.

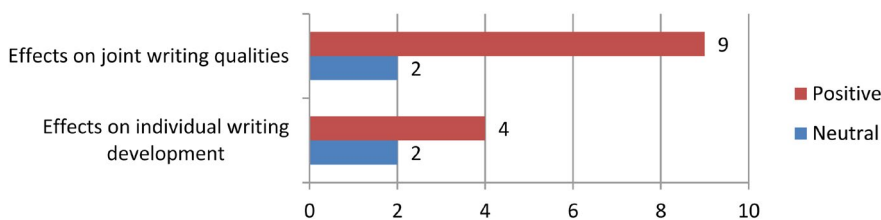


Figure 3. Effects of technologies on students' outcomes of TECW.

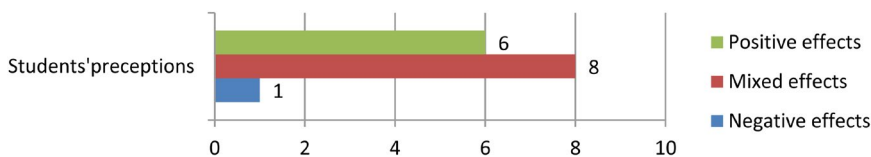


Figure 4. Effects of technologies on students' perceptions of TECW.

4.2. Effectiveness of technologies

4.2.1. Effectiveness of technologies on students' outcomes of TECW

A total of 13 articles investigated students' outcomes of TECW, out of which nine focused on joint writing qualities and four on individual writing development. All 13 studies reported positive effects of technologies on enhancing students' outcomes, while four of them reported neutral effects as well (see Figure 3). More details are presented in Appendix C.

4.2.2. Effectiveness of technologies on students' perceptions of TECW

A total of 15 articles investigated the effects of technologies on students' perceptions of TECW, among which six reported positive results, eight reported mixed results, and one reported negative results (see Figure 4).

4.2.3. Reasons for the effectiveness of technologies

Based on the literature, we identified six main reasons for the effectiveness of technologies in terms of enhancing students' outcomes and perceptions of TECW.

First, technologies enhanced students' outcomes and perceptions of TECW by prompting group interactions. Many studies reported that students' outcomes and perceptions of TECW were positively correlated to their engagement in group interactions (Abrams, 2019; Hsu & Lo, 2018; Mohamadi, 2018; Yeh, 2014), which could be increased by the application of technologies (Aydın & Yıldız, 2014; Bikowski & Vithanage, 2016; Zhang et al., 2014). For example, in Aydın and Yıldız (2014), most students regarded TECW very useful because they found wikis allowed them to extend collaboration outside the classroom and engage in group interactions frequently without any time or space

restrictions. Zhang et al. (2014) reported that their participants devoted more time and effort to the collaboration and communication with their peers when using technology than usual. Woo et al. (2011) and Wang (2014) observed similar situations and explained that it was because of the interactive nature of technologies. Moreover, technologies increased students' engagement in group interactions by reducing their pressure in communication (Chao & Lo, 2011; Wang, 2014, 2015). In traditional face-to-face collaborative writing contexts, students were usually reluctant to make comments or revisions to their peers' writing out of the fear of offending others or embarrassing themselves (Lin & Yang, 2011; Zhang et al., 2014); some students also felt pressure to read comments (Lin & Yang, 2011). However, in technology-enhanced contexts, students felt easy and comfortable in commenting on each other, as reported in Razak and Saeed (2014). Lin and Yang (2011) also found that their students had reduced pressure to read comments with the aid of technology.

Second, technologies enhanced students' outcomes and perceptions of TECW by helping students reflect on their work and identify errors and weakness. The literature has identified significantly positive correlation between students' reflection on work and identification of errors and weakness and their outcomes and perceived usefulness of collaborative writing tasks (Aydın & Yıldız, 2014; Elola & Oskoz, 2010; Hsu & Lo, 2018; Zhang et al., 2014). In TECW, students were allowed a lucid view of their texts, which was hardly available in traditional paper-based contexts (Bikowski & Vithanage, 2016). The lucid view led them to conduct in-depth self-reflection on their work (Bikowski & Vithanage, 2016; Elola & Oskoz, 2010; Hsu & Lo, 2018) and facilitated them to identify their errors and weakness in writing (Aydın & Yıldız, 2014; Zhang et al., 2014). For example, Aydın and Yıldız (2014), Elola and Oskoz (2010), and Wang (2015) reported that technologies drew their students' attention to the form of writing and made their grammatical mistakes highly noticeable, which "pushed" their students to reflect on the grammatical accuracy of their works and their prior knowledge of grammar. Also, Hsu and Lo (2018) reported that the use of wikis raised their participants' awareness of idea expression and encouraged them to reflect on how their ideas were developed, whether their ideas were relevant to the topics, and how their expressions could convey their ideas adequately and accurately. Moreover, with the aid of technologies, students were more likely to notice their peers' mistakes and encourage each other to reflect on their writing (Abrams, 2019; Elola & Oskoz, 2010; Wang, 2014). This type of peer review fostered students' awareness of their weaknesses in the target language (Wang, 2014).

Third, technologies enhanced students' outcomes and perceptions of TECW by boosting learner motivation and confidence. The literature reported that students tended to feel more motivated in TECW than in traditional contexts because "the active and innovative nature" (Wang, 2014, p. 388) of technologies rendered the writing tasks to be more interesting and engaging (Wang, 2014;

Zorko, 2009). For example, Zhang et al. (2014) reported that most of their participants regarded blogging as an attractive and motivating approach to TECW. Furthermore, technologies afforded the students to receive frequent feedback and attention from their peers, which was also a source of motivation (Woo et al., 2011; Zhang et al., 2014). As supporting evidence, Woo et al. (2011) quoted their participants, “We can use the Wiki-like an MSN to talk” and “They[My group members] give me some messages, and I feel so happy.” In addition, technologies recorded all versions of writing and made students’ progression and development very noticeable, which helped students cultivate a sense of achievement (Lin & Yang, 2011; Wang, 2014) and confidence (Aydin & Yildiz, 2014; Bikowski & Vithanage, 2016; Wang, 2014, 2015). The enhanced motivation and confidence enabled students to achieve satisfying outcomes and positive perceptions of this TECW (Bikowski & Vithanage, 2016; Lin & Yang, 2011; Wang, 2014). For example, Ducate et al. (2011) and Wang (2014) reported that technologies improved their students’ motivation to use the target language and confidence in L2 writing and thereby enabled them to held positive attitudes toward the collaborative learning tasks. Bikowski and Vithanage (2016) also observed these positive effects of technologies and argued that the enhanced motivation and confidence could be transferred to individual writing activities and improve students’ individual writing abilities.

Fourth, technologies enhanced students’ outcomes and perceptions of TECW by facilitating writing. Technologies afforded multiple students to write without constraint on time or space (Bikowski & Vithanage, 2016; Wang, 2014, 2015; Wong et al., 2011), so engagement in co-construction of text was more accessible in TECW than in traditional contexts. Thus, technologies increased students’ opportunities to perform writing practices (Aydin & Yildiz, 2014; Chao & Lo, 2011; Lin & Yang, 2011). For example, Chao and Lo (2011) reported that their participants engaged in writing “more continuously and enthusiastically” in TECW than usual because of the support of various editing functions of technologies (p. 397). Similarly, Woo et al. (2013) observed the usefulness of technologies in increasing students’ writing practice, especially revising, and argued that the increased revising would result in enhanced joint writing qualities. Furthermore, the powerful functionalities for editing and revising facilitated writing practices that used to be hardly available in traditional contexts, such as planning and drafting, which also led to enhanced outcomes and perceptions of collaborative writing (Bikowski & Vithanage, 2016; Elola & Oskoz, 2010). For example, Elola and Oskoz (2010) found that their participants used technologies to plan and draft the structure and organisation of their writing at the beginning and then improved them by switching, expanding, polishing, and deleting paragraphs and sentences with the aid of various functions of wikis. The researchers argued that such convenience of “playing with” paragraphs and sentences was impossible in paper-based drafting and resulted in the significant improvement of their participants’ joint writing in

structure and organisation. Bikowski and Vithanage (2016) identified similar results in their investigation of the effects of Google Docs on TECW. As they reported, technology brought about a “new way” of writing that helped their students plan, proofread, and revise their writing in ways unavailable in paper-based context, which led to their enhanced writing proficiency.

Fifth, technologies enhanced students’ outcomes and perceptions of TECW by encouraging students to learn from others. Technologies afforded students to view all changes in text by other members and outputs of other groups, which was unavailable in traditional paper-based contexts (Li, 2013; Li & Zhu, 2016, 2017). This advantage resulted in enhanced outcomes and perceptions of TECW because it raised enabled students to compare their works and changes to those of other groups or members (Zorko, 2009), receive new information and knowledge (Hsu & Lo, 2018; Lin & Yang, 2011), and learn from others (Lin & Yang, 2011; Woo et al., 2011; Zorko, 2009). For example, Lin and Yang (2011) found that their participants took advantage of the neat organisation and automatic history recording of wikis to proactively read the changes made by other members and the works of other groups. Thereby, the participants learned from others and achieved significant progress in spelling, grammar, style, and quality of writing within a short time. Other researchers reported that students benefited from reading others’ works in their development of writing content (Elola & Oskoz, 2010; Wang, 2015), structure (Elola & Oskoz, 2010; Li & Zhu, 2017), coherence (Li & Zhu, 2017), and language style (Wang, 2015).

Sixth, technologies enhanced students’ outcomes and perceptions of TECW by making the learning process enjoyable. Many articles reported that students had a better learning experience in TECW than in traditional writing tasks (Chao & Lo, 2011; Wong et al., 2011; Woo et al., 2011). As revealed by the results of Ducate et al.’s (2011) questionnaire survey, the majority of students indicated that they liked the TECW project and enjoyed working with technology. The results of Aydın and Yıldız’s (2014), Chao and Lo (2011), and Wong et al.’s (2011) interviews also suggested that the students enjoyed working collaboratively on the wiki-pages. Such enjoyable learning experience resulted in students’ enhanced outcomes and positive perceptions of TECW (Chao & Lo, 2011; Ducate et al., 2011; Woo et al., 2011). The positive learning experiences were closely related to the sense of freshness brought by the innovation and attraction of technologies (Wang, 2014; Zhang et al., 2014). Zhang et al. (2014) also reported that their students enjoyed using the technologies that were related to their daily life, especially social life, such as blog (sina.com) in their case.

4.2.4. Conditions that may lead to ineffective use of technology

Several articles reported neutral or negative results in their investigation of the effects of technologies on students’ outcomes and perceptions of TECW,

of which we identified three main conditions that may lead to ineffective use of technology.

First, technologies may be ineffective in enhancing students' outcomes and perceptions of TECW when technologies were inconvenient to use. Many researchers, such as Ducate et al. (2011), Elola and Oskoz (2010), and Wang (2014), reported that their participants encountered technical problems during the process of TECW. For example, Wang (2014) found that some students complained about the text formatting and browser compatibility of wikis and felt discouraged in using technologies. Li and Zhu's (2013) participants regarded the prerequisites for computers and Internet in TECW limited their engagement and reduced their motivation in learning. Chao and Lo (2011) noted that internet connection problems occurred very frequently and largely reduced students' efficiency and perceptions of TECW. Woo et al.'s (2011) participants reported that it sometimes took them over 30 min to load documents using PBworks, which caused that they partially abandoned this tool and used personal USBs and emails to share documents. Zorko (2009) also found that the participants preferred Messenger and emails in group interactions to wikis.

Second, technologies may be ineffective in terms of enhancing students' outcomes and perceptions of TECW when students did not devote adequate effort to planning and reflection. Although technologies could usually facilitate students to plan and reflect on their writing (Bikowski & Vithanage, 2016; Elola & Oskoz, 2010; Hsu & Lo, 2018), some students paid limited effort on these two stages of TECW, which reduced the effectiveness of technologies in their writing development and their perceived usefulness of the writing activities. For example, Hsu and Lo (2018) reported that some students tended to skip the planning and outlining stage at the beginning of writing even though wikis, the tool that they used, provided powerful functions for this stage. Such stage skipping resulted in little improvement of structure and organization in students' individual writing. Bikowski and Vithanage (2016) found that most of their students failed to perceive the usefulness of TECW although they did achieve overall development in individual writing ability. The researchers explained that it was because students did not reflect on their process of writing or their development as writers and collaborators even though Google Docs, the technology they used, supported self-reflection.

Third, technologies may be ineffective in enhancing students' outcomes and perceptions of TECW when students lacked co-responsibility for the writing. Li and Zhu (2017) identified this situation in one group of their participants who failed to reach satisfying collaboration or good outcomes of TECW. Although the students easily identified the errors of other group members with the aid of wiki features, the members hardly responded to the identification. The researchers argued that the lack of co-responsibility diminished peer scaffolding and negatively influence the wiki joint writing.

5. Conclusion and discussion

This study presents an in-depth review of the empirical studies on TECW published from 2009 to 2019, illustrating the main types, features, and effectiveness of technologies in TECW.

5.1. Implications for educators

Based on the results of the current review, we presented two main pedagogical implications for teachers who were interested in TECW.

First, teachers should select proper technologies. Despite the overall effectiveness of technologies as found in this review, these technologies varied in features and functions, as shown in the results of this review. Thus, teachers should carefully select technologies for TECW at the beginning (Zhang et al., 2014; Zorko, 2009). The user-friendliness of technology should be taken into the consideration because it was closely related to students' perceptions of TECW (Chao & Lo, 2011; Ducate et al., 2011). Wikis were recommended due to their being easy-to-learn and easy-to-use (Chao & Lo, 2011; Wang, 2014, 2015). Google Docs were regarded to be even more usable and steady than wikis by Abrams (2019). Technologies that were closely related to daily social life and could be downloaded as mobile applications, such as Facebook and blog (sina.com) were usually familiar and attractive to students and may thereby be helpful in TECW (Selcuk et al., 2019; Zhang et al., 2014). Moreover, teachers should consider the target language they were going to teach. If the use of special characters were needed in the L2, the use of Google Docs were recommended (Abrams, 2016). Types of writing tasks may also be considered in the selection of technology. For example, the non-linear navigation of wikis may be suitable for creative writing, such as story branching (Ducate et al., 2011), while informative writing may be well-supported by Google Docs which afforded the attachment of tables (Abrams, 2016). Teachers should also consider which types of communications they wish their students have in TECW with awareness of the fact that wikis supported asynchronous interactions (Bradley et al., 2010; Li, 2013), Google Docs (Abrams, 2016, 2019), Facebook (Razak & Saeed, 2014; Selcuk et al., 2019), and chats (Cho, 2017; Elola & Oskoz, 2010) supported synchronous interactions, wikis (Wong et al., 2011), and Google Docs (Kessler et al., 2012), and forums (Mohamadi, 2018) supported on-line commentary. If teachers wish to provide students with learning materials during the process of TECW, wikis (Bradley et al., 2010; Woo et al., 2011) and Facebook (Razak & Saeed, 2014; Selcuk et al., 2019) were recommended.

Second, teachers should assess students' technical literacy and, according to the results of assessment, provide them with instruction and training on technology. Although most technologies we reviewed were labelled as "easy-to-learn"

(Chao & Lo, 2011; Wang, 2014, 2015), it may be inappropriate to assume that all students were able to use technologies smoothly and effectively because many researchers reported that their participants encountered difficulty in using technologies (Chao & Lo, 2011; Ducate et al., 2011; Elola & Oskoz, 2010; Wang, 2014). Therefore, it may be helpful to assess students' technical literacy like Kessler (2009) did. Kessler required the students to complete an online orientation to the course management system, screen-recorded their mouse clicks during the process, and analysed their performance. Based on the analyses, the researcher assessed students' abilities to access online recourses and use digital tools. If the students demonstrated difficulty using technologies, teachers should provide them with instruction and training on technology before the implementation of TECW (Ducate et al., 2011; Kessler, 2009; Li, 2013; Li & Zhu, 2017). The instruction should cover the introductions to the features and modules of technology useful in TECW (Li, 2013; Li & Zhu, 2017), the explanations of the usefulness of these features and modules in TECW (Li, 2013; Li & Zhu, 2017), and the methods and examples of using these features and modules (Li & Zhu, 2013; Zorko, 2009). Practicing the use of technology was also needed to make students fully prepared for TECW (Chao & Lo, 2011; Ducate et al., 2011).

Third, teachers should provide students with explicit instruction on TECW. As shown in the results of this review, technologies afforded many "new" practices of collaborative writing that were hardly available in traditional face-to-face contexts, such as convenient planning, drafting, and revising (Bikowski & Vithanage, 2016; Elola & Oskoz, 2010), comprehensive reflection on work (Elola & Oskoz, 2010; Hsu & Lo, 2018), quick identification of errors (Aydın & Yıldız, 2014; Zhang et al., 2014), and easy learning from previous versions of writing, changes by other members, and work of other groups (Li, 2013; Li & Zhu, 2016, 2017). By those technology-enhanced practices, students may achieve enhanced outcomes and perceptions of TECW, as shown in the results of this review. However, the affordances for "new" practices of technologies did not automatically lead to students' performance of those practices (Woo et al., 2013). Some students may not be fully aware or capable of these practices and continued using traditional collaborative writing methods in TECW (Bikowski & Vithanage, 2016; Li & Zhu, 2017; Lin & Yang, 2011). The lack of those practices may result in the ineffectiveness of technologies and reduce students' perceived usefulness of technologies (Hsu & Lo, 2018; Bikowski & Vithanage, 2016; Li & Zhu, 2017). Therefore, it was valuable to provide students with explicit instruction on how to perform collaborative writing in technology-enhanced environments, so students' awareness and capability of writing and collaboration could evolve in pace with the development of technology, consistent with Cho (2017), Kessler et al. (2012), and Woo et al. (2013). For example, Woo et al. (2013) taught their participants how to make meaningful revision in TECW by taking full advantages of the wiki features. Kessler et al. (2012) led their students to consider series of issues related to the performance of TECW, such as how many comments they

should provide to each other, how much writing could be completed online, and how much should be completed when all group members were sitting physically together. Bikowski and Vithanage (2016) helped their students conduct self-reflection by raising their awareness of writing needs and encouraging them to compare their writing to those of their peers.

5.2. Implications for researchers

The results of our study have some similarities with the previous review studies. For example, we found that wikis and Google Docs were investigated very frequently in TECW studies, consistent with Li (2018) and Storch (2019). This similarity indicated the constant effectiveness and technical maturity of wikis and Google Docs in supporting collaborative activities for L2 learning. Future researchers and educators may use these two types of technologies in TECW to enhance L2 learning.

In addition to the similarities, this review also identifies several new tendencies and developments in the field of TECW. First, Facebook and chats were frequently used in collaborative writing (e.g., in Cho, 2017; Elola & Oskoz, 2010; Razak & Saeed, 2014), while previous review studies on TECW hardly examined these two types of technologies. Empirical evidence was found supporting the usefulness of Facebook and chats in enhancing students' outcomes and perceptions of TECW (e.g., de Larreta-Azelain, 2013; Hattem, 2014), which entailed the value of further investigation and application of them in TECW.

Second, most previous reviews regarded the technology-enhanced collaborative writing equal to the computer-mediated collaborative writing (e.g., Li, 2018; Storch, 2011, 2019) but ignored other devices that were potentially useful in this learning activity. Nonetheless, this review found that mobile devices were usable and well-perceived in collaborative writing activities (e.g., in Selcuk et al., 2019). Moreover, many types of technologies that we identified as useful in TECW were also usable on mobile devices, such as Google Docs, Facebook, and Twitter, indicating the considerable potential of mobile devices as the mediation of collaborative writing. Future researchers may conduct further investigation on the use of mobile devices in TECW.

Third, this review reported several cases of TECW without Internet support, where students engaged in collaborative writing based on word processors and face-to-face interactions (e.g., Van Steendam et al., 2014; Wang, 2015; Zhang, 2018), while most previous reviews assumed the constant accessibility of Internet service throughout the process of TECW (e.g., Godwin-Jones, 2003; Li, 2018; Storch, 2019). As empirical evidence was found supporting the feasibility and effectiveness of offline TECW in improving students' writing outcomes of TECW (e.g., in Wang, 2015; Zhang, 2018), this study suggests that researchers further investigate and implement TECW based on word processors and face-to-face interactions.

5.3. Future use of technologies in TECW

Recent years have witnessed the increasing integration of technologies to L2 education (Chen et al., 2019; Shadiev et al., 2017; Shadiev & Yang, 2020; Zhang & Zou, 2020). Some technologies frequently used in TECW have also been investigated in other learning activities. For instance, Facebook can also be used in the flipped L2 classrooms to increase students' opportunities for oral skill practices (Lin & Hwang, 2018; Zou et al., 2020).

However, some technologies have yet been investigated in TECW that had been widely used and demonstrated potential in other language learning activities. Augmented reality (AR) technology was an example. In language educational contexts, AR may superimpose language instructions into real-life contexts (Chang, Hou, Pan, Sung, & Chang, 2015; Chen et al., 2020), facilitate students' language input and output, encourage peer-to-peer interactions (Ho, Hsieh, Sun, & Chen, 2017; Huang & Chuang, 2016), and thereby be potentially helpful in enhancing students' writing. Thus, researchers and educators may investigate and apply AR technology in their future TECW projects. Lin, Liu, and Chen's (2020) study on enhancing writing activities with AR technology may be an example to follow. Adapted from Lin et al.'s (2020) research design, researchers and educators of TECW may require their students to have a museum tour in small groups. During the tour, students used tablets to collect and view AR-enhanced learning materials presented in the museum and complete collaborative writing tasks with Bluetooth portable keyboards. Since Lin et al. (2020) reported positive results of their AR-enhanced writing activity, it was promising that the application of AR in TECW could also result in students' satisfying writing outcomes and positive perceptions.

5.4. Limitations

There are two limitations of the present review study. First, the articles we reviewed were limited to those published in SSCI journals. Future research might expand the source of studies and include book chapters, conference papers, and journal articles indexed by AHCI, Scopus and ERIC, etc., so as to provide a more comprehensive review in this field. Second, this study focused on the types, features, and effectiveness of educational technologies in TECW, so we did not discuss other aspects of the review articles, such as participants' ages, participants' language proficiency levels, group sizes, and research methods. To fill this gap, future study may review the TECW articles from a wide range of aspects.

Disclosure statement

No potential conflict of interest was reported by the authors.

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Appendix B. Technologies and features useful in TECW

Table B1. Technologies and features useful for students in TECW.

Types of technology	Features	Authors
Wikis	Convenient writing and revising	Bradley et al. (2010), Chao & Lo (2011), Elola & Oskoz (2010), Li (2013), Li & Zhu (2017), Wang (2014), Wang (2015), Wong et al. (2011)
	Convenient group discussions	Bradley et al. (2010), Li (2013), Zorko (2009), Woo et al. (2011)
	User-friendliness	Chao & Lo (2011), Ducate et al. (2011), Wang (2014), Wang (2015)
	Learning materials and tools	Bradley et al. (2010), Woo et al. (2011), Zorko (2009)
	Multimedia technologies	Woo et al. (2011), Woo et al. (2013)
	Personalised scaffolding	Li (2013), Zorko (2009)
Google Docs	Convenient group discussions	Abrams (2016), Abrams (2019), Bikowski & Vithanage (2016), Cho (2017), Kessler et al. (2012)
	Convenient writing and revising	Bikowski & Vithanage (2016), Kessler et al. (2012)
	Availability of special characters, tables, and hyperlinks	Abrams (2016), Abrams (2019)
	User-friendliness	Abrams (2016), Abrams (2019)
	Convenient writing	Abrams (2016), Kessler et al. (2012)
Offline word processor	Multimedia technologies	Abrams (2019)
	n/a	Van Steendam et al. (2014), Wang (2015), Zhang (2018)
Chat	Convenient group discussions	Cho (2017), Elola & Oskoz (2010)
Facebook	Learning materials	Razak & Saeed (2014), Selcuk et al. (2019)
	Convenient group discussions	Razak & Saeed (2014)
	Availability on various devices, such as computers and smartphones	Selcuk et al. (2019)
Forum	User-friendliness	Selcuk et al. (2019)
	Convenient group discussions	Levrai & Bolster (2019), Mohamadi (2018)
Group Scribbles	Convenient revising	Mohamadi (2018)
	Convenient writing	Wen et al. (2015)
EtherPad	Convenient group discussions	Wen et al. (2015)
	Convenient writing and revising	Yeh (2014)
Blog	Convenient group discussions	Yeh (2014)
	Convenient group discussions	Zhang et al. (2014)
	User-friendliness	Zhang et al. (2014)

Table B2. Technologies and features useful for researchers and educators in TECW.

Types of technology	Features of technologies	Authors
Wikis	Automatic saving, organising, and analysis of historical records of the process of TECW	Aydin & Yildiz (2014), Bradley et al. (2010), Ducate et al. (2011), Elola & Oskoz (2010), Hsu & Lo (2018), Kessler (2009), Lai et al. (2016), Li (2013), Li & Kim (2016), Li & Zhu (2013), Li & Zhu (2016), Li & Zhu (2017), Wang (2015), Wong et al. (2011), Woo et al. (2011), Woo et al. (2013), Zorko (2009)
	Group setting	Aydin & Yildiz (2014), Bradley et al. (2010), Chao & Lo (2011), Elola & Oskoz (2010), Kessler (2009), Li & Kim (2016), Li & Zhu (2013), Wang (2014), Wang (2015), Woo et al. (2011), Woo et al. (2013), Zorko (2009)
	Feedback giving	Elola & Oskoz (2010), Woo et al. (2011), Woo et al. (2013), Zorko (2009)
	Simplicity to incorporate into a course	Wang (2014), Wang (2015)
Google Docs	Automatic saving and organising of historical records of the process of TECW	Abrams (2019), Bikowski & Vithanage (2016), Cho (2017), Kessler et al. (2012)
	Group setting	Abrams (2019)
Offline word processor	n/a	Van Steendam et al. (2014), Wang (2015), Zhang (2018)
Chat	Automatic saving of historical records of the communication process of TECW	Cho (2017), Elola & Oskoz (2010)
Facebook	Automatic saving of historical records of the communication process of TECW	Razak & Saeed (2014), Selcuk et al. (2019)
Forum	Feedback giving	Levrai & Bolster (2019), Mohamadi (2018)
Group Scribbles	Group setting	Wen et al. (2015)
EtherPad	Automatic saving and organising of historical records of the process of TECW	Yeh (2014)
Stormboard	Automatic saving of historical records of the communication process of TECW	Levrai & Bolster (2019)
Interactive Whiteboard	Visualisation and monitoring of the communication process of TECW	Wen et al. (2015)

Appendix C. Effectiveness of technologies in TECW

Table C1. Effectiveness of technologies in TECW on the joint writing quality.

Results	Reasons	Authors
Technologies had positive effects on joint writing qualities.	Technologies prompted group interactions.	Abrams (2019), Ducate et al. (2011), Li & Zhu (2017), Mohamadi (2018), Yeh (2014)
	Technologies helped students reflect on their work and identify their errors and weakness in writing.	Abrams (2019), Aydın & Yıldız (2014), Elola & Oskoz (2010)
	Technologies facilitated writing.	Elola & Oskoz (2010), Woo et al. (2013)
	Technologies eliminated students' cognitive load in writing.	Elola & Oskoz (2010)
	Technologies helped students foster learning autonomy.	Mohamadi (2018)
Technologies had neutral effects on joint writing qualities.	The treatment session was too short.	Elola & Oskoz (2010)
	There was a lack of collective cognition and co-ownership among group members in TECW.	Li & Zhu (2017)

Table C2. Effectiveness of technologies in TECW on the individual writing development.

Results	Reasons	Authors
Technologies had positive effects on individual writing development.	Technologies prompted group interactions.	Bikowski & Vithanage (2016), Hsu & Lo (2018), Wang (2015), Wong et al. (2011)
	Technologies helped students reflect on their work and identify their errors and weakness in writing.	Bikowski & Vithanage (2016), Hsu & Lo (2018), Wang (2015), Wong et al. (2011)
	Technologies facilitated writing.	Bikowski & Vithanage (2016), Hsu & Lo (2018), Wang (2015)
	Technologies boosted learner motivation and confidence.	Bikowski & Vithanage (2016)
	Technologies enabled students to receive new information.	Wang (2015)
	Technologies eliminated students' cognitive load.	Hsu & Lo (2018)
Technologies had neutral effects on individual writing development.	The treatment session was too short.	Hsu & Lo (2018)
	Students did not spend sufficient time planning and drafting their writing.	Hsu & Lo (2018)

Table C3. Effectiveness of technologies on students' perceptions of TECW.

Results	Reasons	Authors
Technologies had positive effects on students' perceptions of TECW.	Technologies boosted learner motivation and confidence.	Aydin & Yıldız (2014), Bikowski & Vithanage (2016), Bradley et al. (2010), Chao & Lo (2011), Ducate et al. (2011), Elola & Oskoz (2010), Lin & Yang (2011), Wang (2014), Wang (2015), Wong et al. (2011), Woo et al. (2011), Zhang et al. (2014), Zorko (2009)
	Technologies prompted group interactions.	Chao & Lo (2011), Elola & Oskoz (2010), Lin & Yang (2011), Wang (2014), Woo et al. (2011), Zhang et al. (2014), Zorko (2009)
	Technologies improved learning experience.	Aydin & Yıldız (2014), Ducate et al. (2011), Wang (2014), Wang (2015), Wong et al. (2011), Woo et al. (2011), Zhang et al. (2014)
	Technologies facilitated writing.	Aydin & Yıldız (2014), Chao & Lo (2011), Elola & Oskoz (2010), Lin & Yang (2011)
	Technologies helped students reflect on their work and identify their errors and weakness in writing.	Elola & Oskoz (2010), Lin & Yang (2011), Wang (2014), Zhang et al. (2014)
	Technologies were innovative and easy-to-use.	Lin & Yang (2011), Wang (2014)
	Technologies provided student with a context for knowledge application.	Ducate et al. (2011), Wang (2014)
	Technologies enabled students to receive immediate feedback from their teachers.	Lin & Yang (2011)
	Technologies increased learning opportunities.	Lin & Yang (2011)
	Technologies were inconvenient to use.	Chao and Lo (2011), Ducate et al. (2011), Elola & Oskoz (2010), Li & Zhu (2013), Lin & Yang (2011), Wang (2014), Woo et al. (2011), Zorko (2009)
Technologies had negative effects on students' perceptions of TECW.	Students were reluctant to learn and use new technologies.	Lin & Yang (2011)
	Students did not reflect on TECW.	Bikowski & Vithanage (2016)